

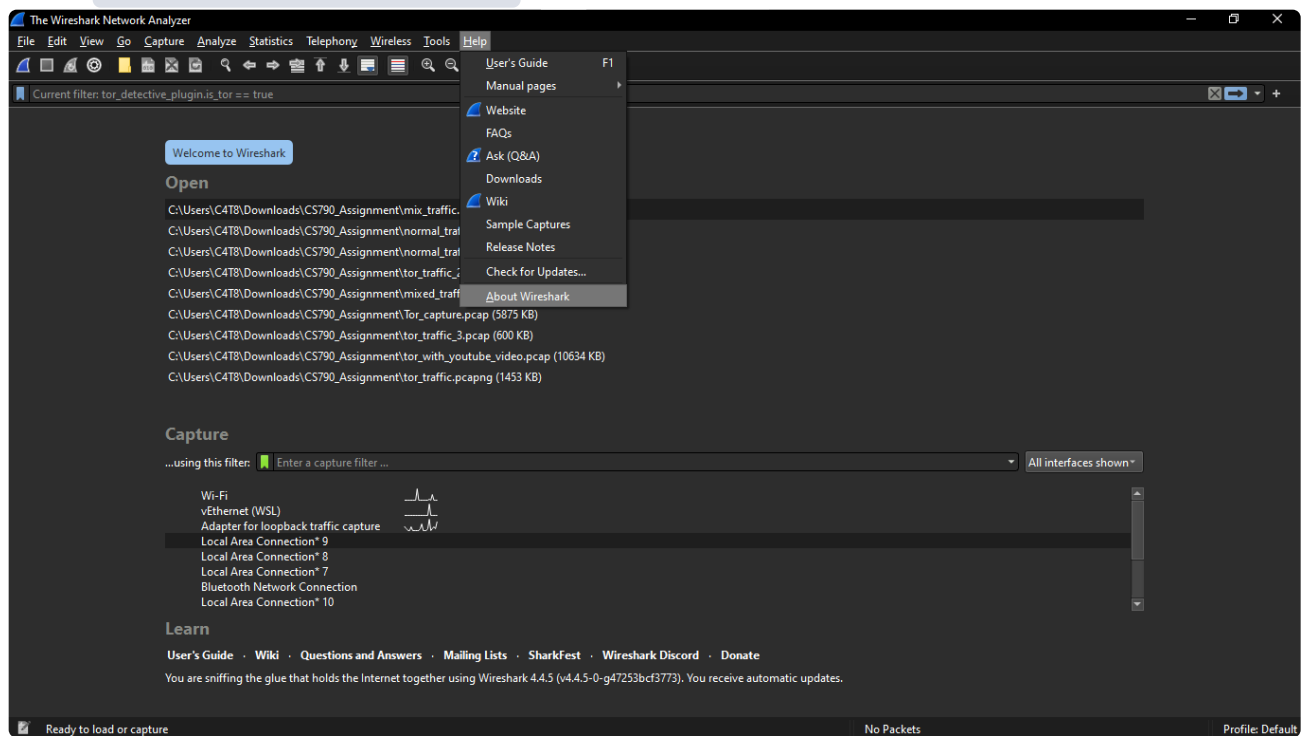
CS790 Assignment 2 README

Name: Vansil Rakeshkumar Chauhan (24m0847)

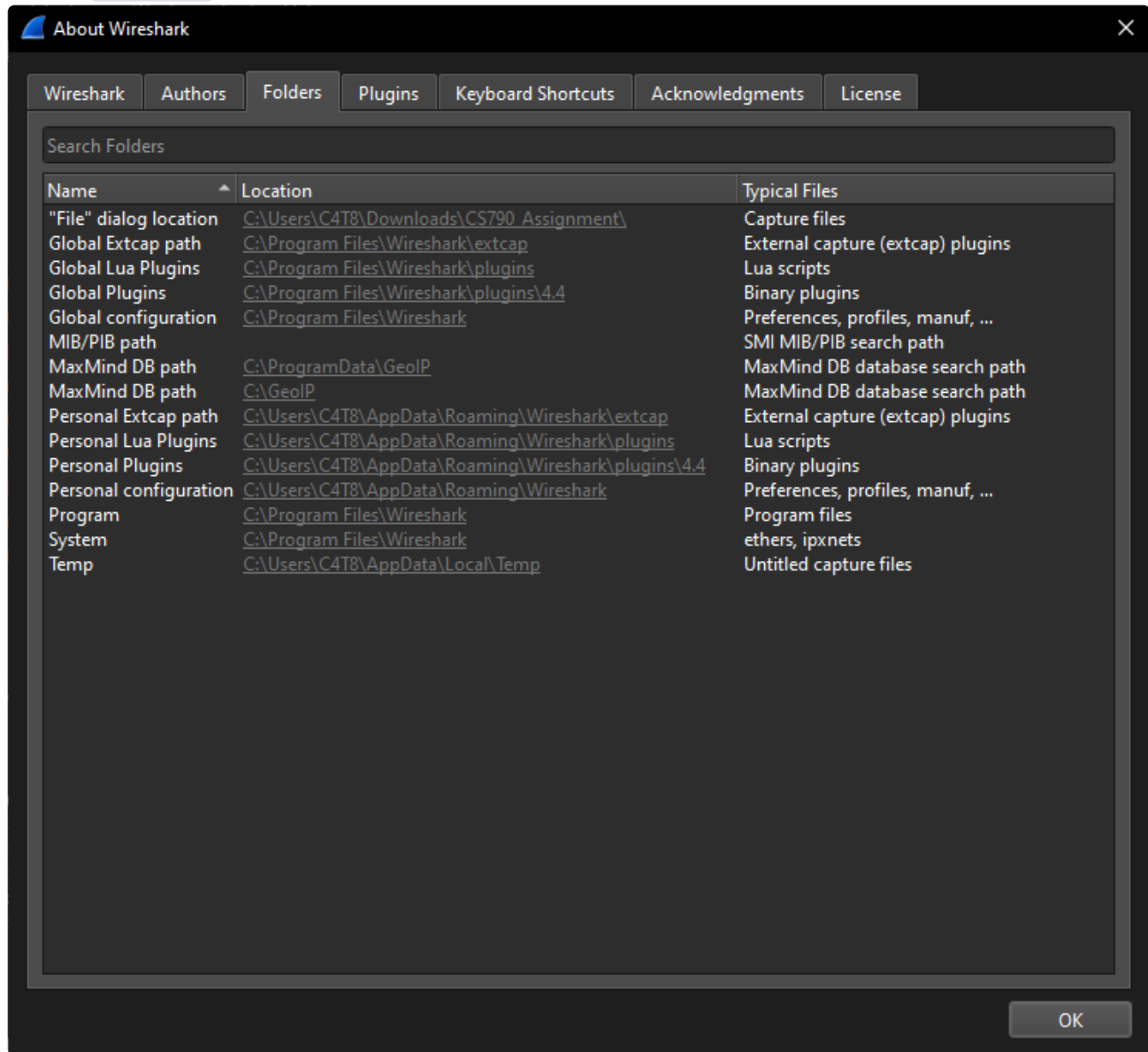
Title: The Evil Eye

Install a lua plugin

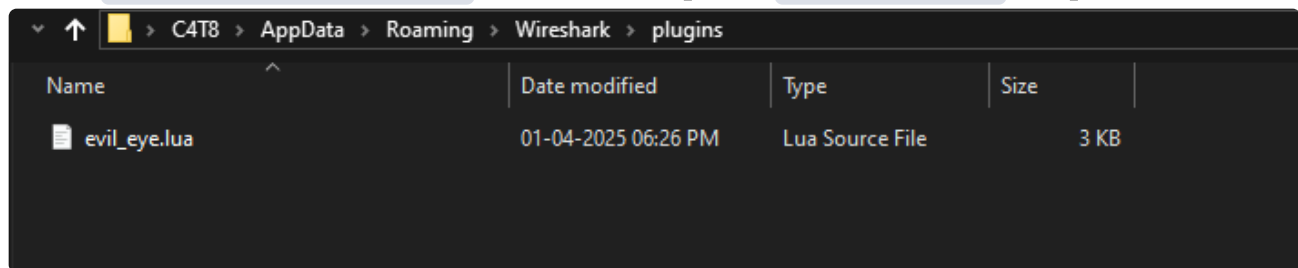
1. Open Wireshark
2. Click Help → About Wireshark



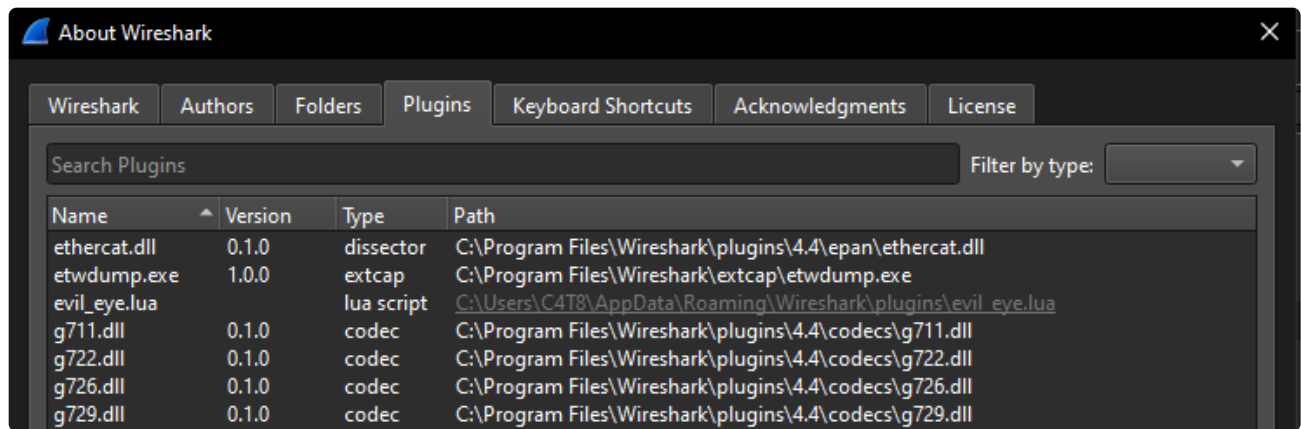
3. Go to **Folders** tab.



4. Go to **Personal Lua Plugins** location and paste **evil_eye.lua** script there.

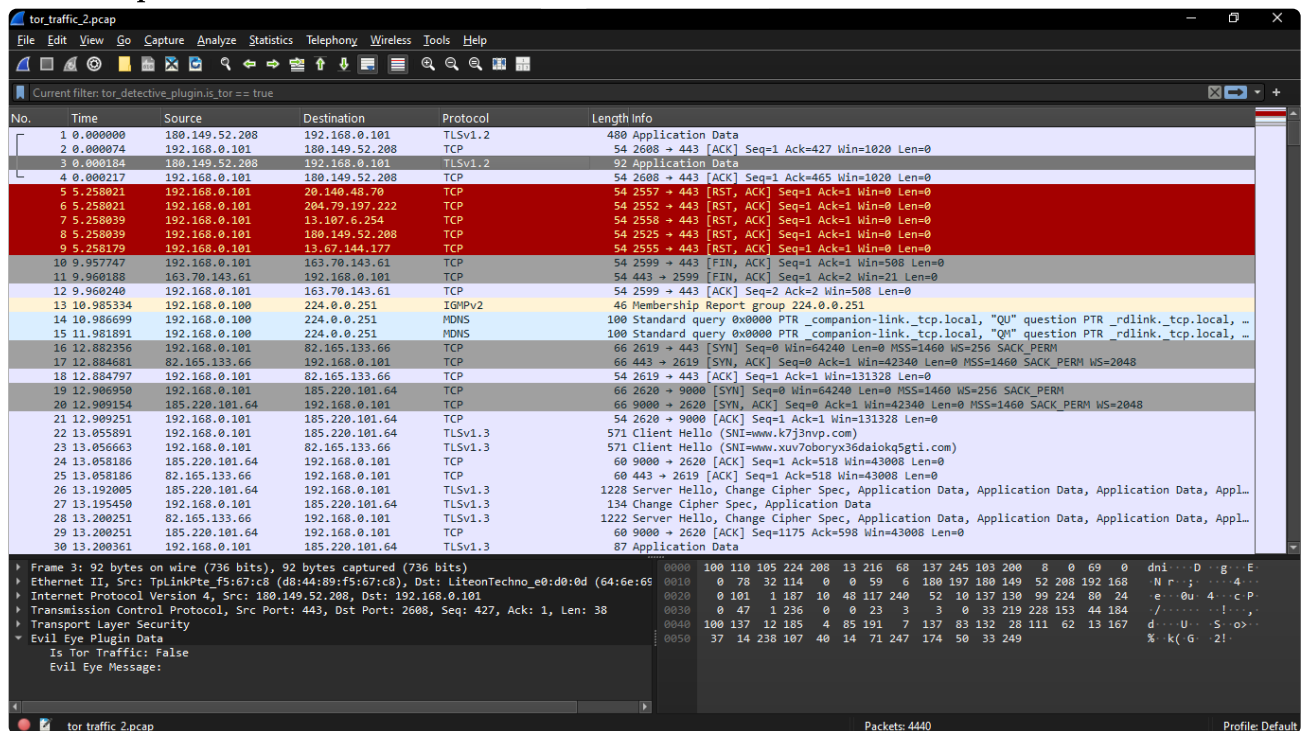


5. Restart **Wireshark** or press **ctrl+shift+L** to reload **lua** plugins. It should be visible under **Help → About Wireshark → Plugins**



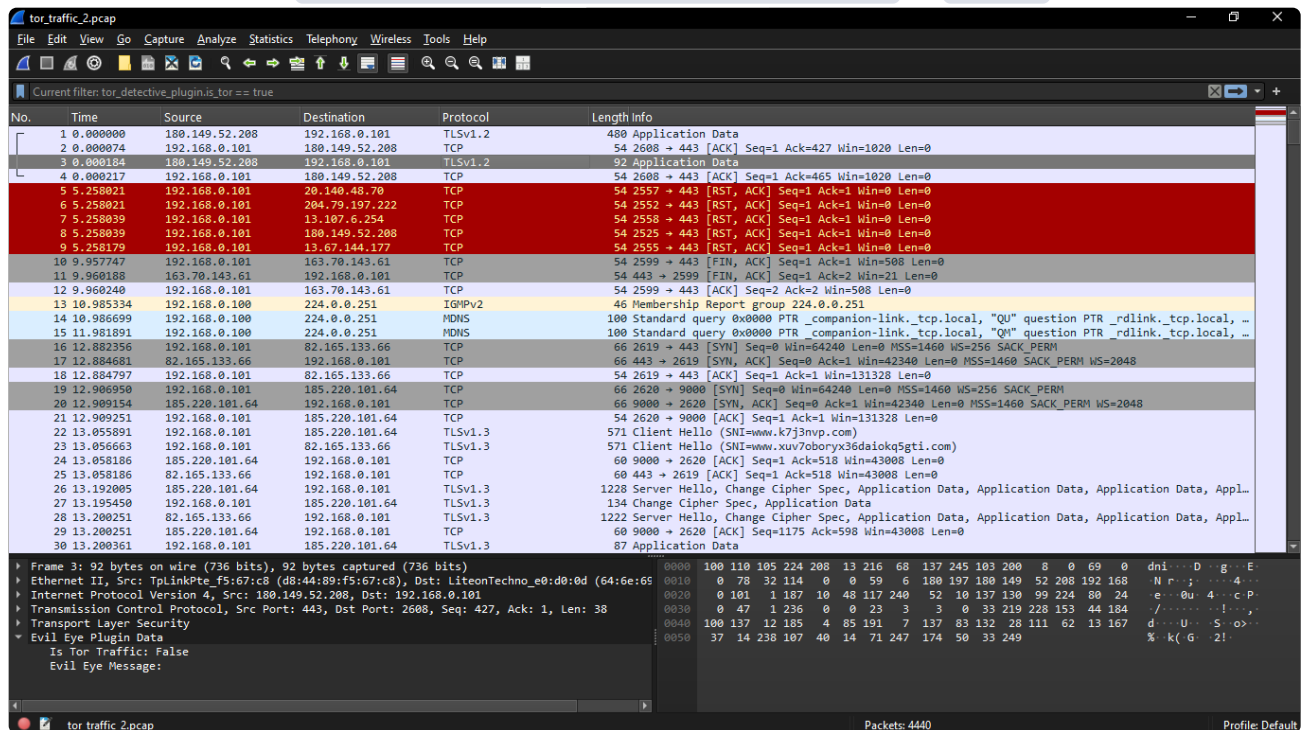
Check if plugin is working

1. Open .pcap file which contains TOR traffic in it. or Run tor browser and capture from respective interface. (ex. WiFi)

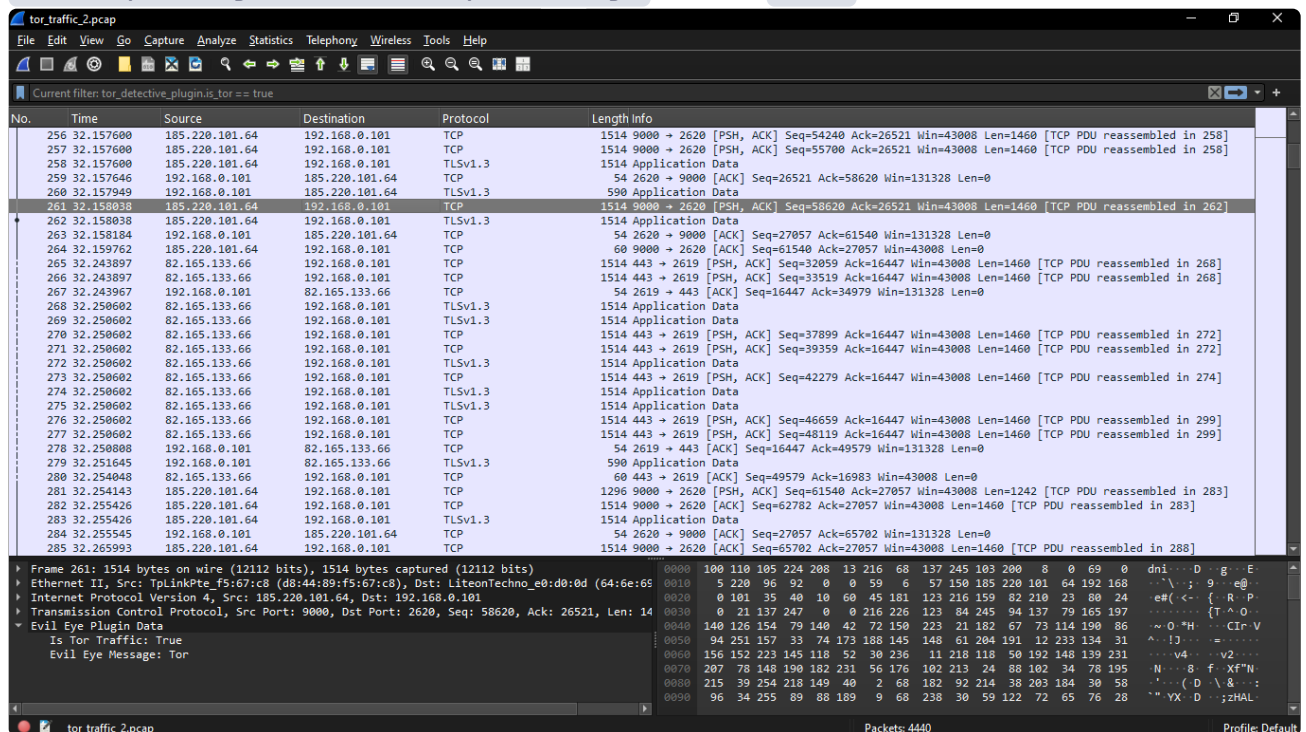


2. You should be able to see Evil Eye Plugin Data subtree.

3. For normal traffic, Evil Eye Plugin → Is Tor Traffic is False

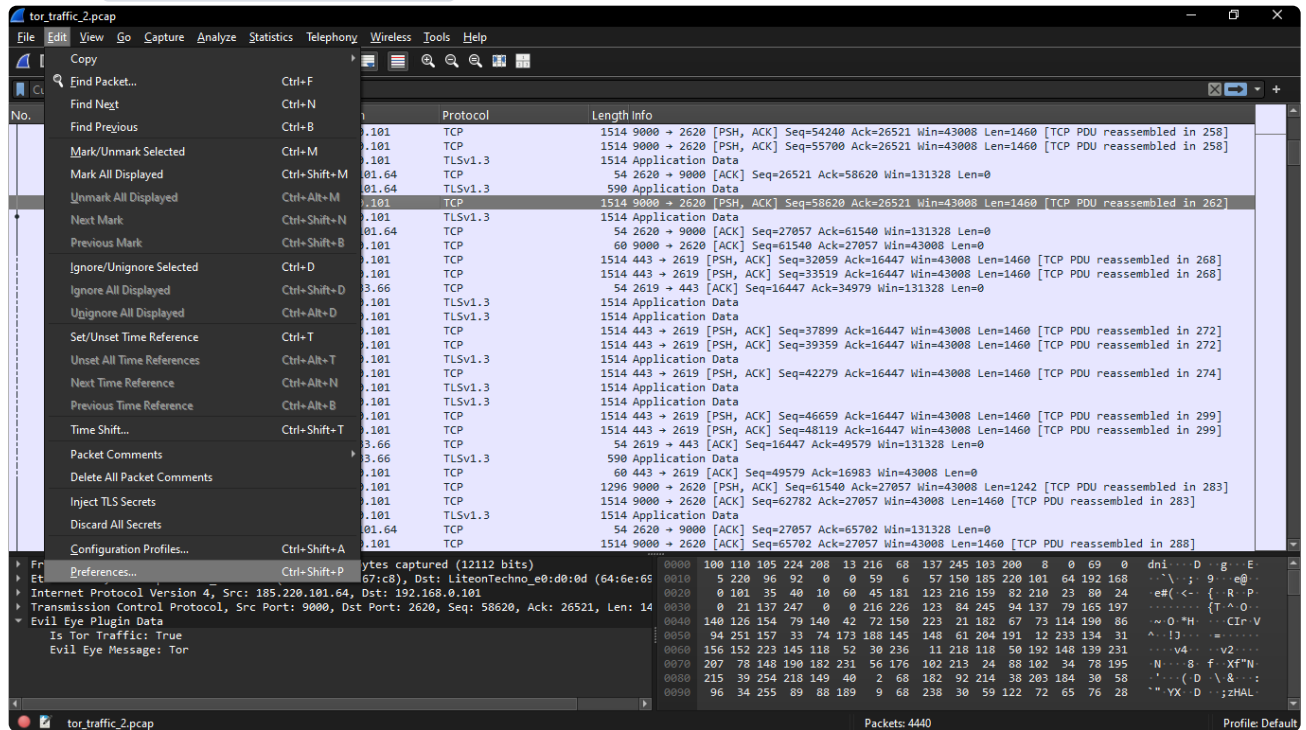


4. For suspected TOR traffic, Evil Eye Plugin → Is Tor Traffic will be True and Evil Eye Plugin → Evil Eye Message will be Tor!.

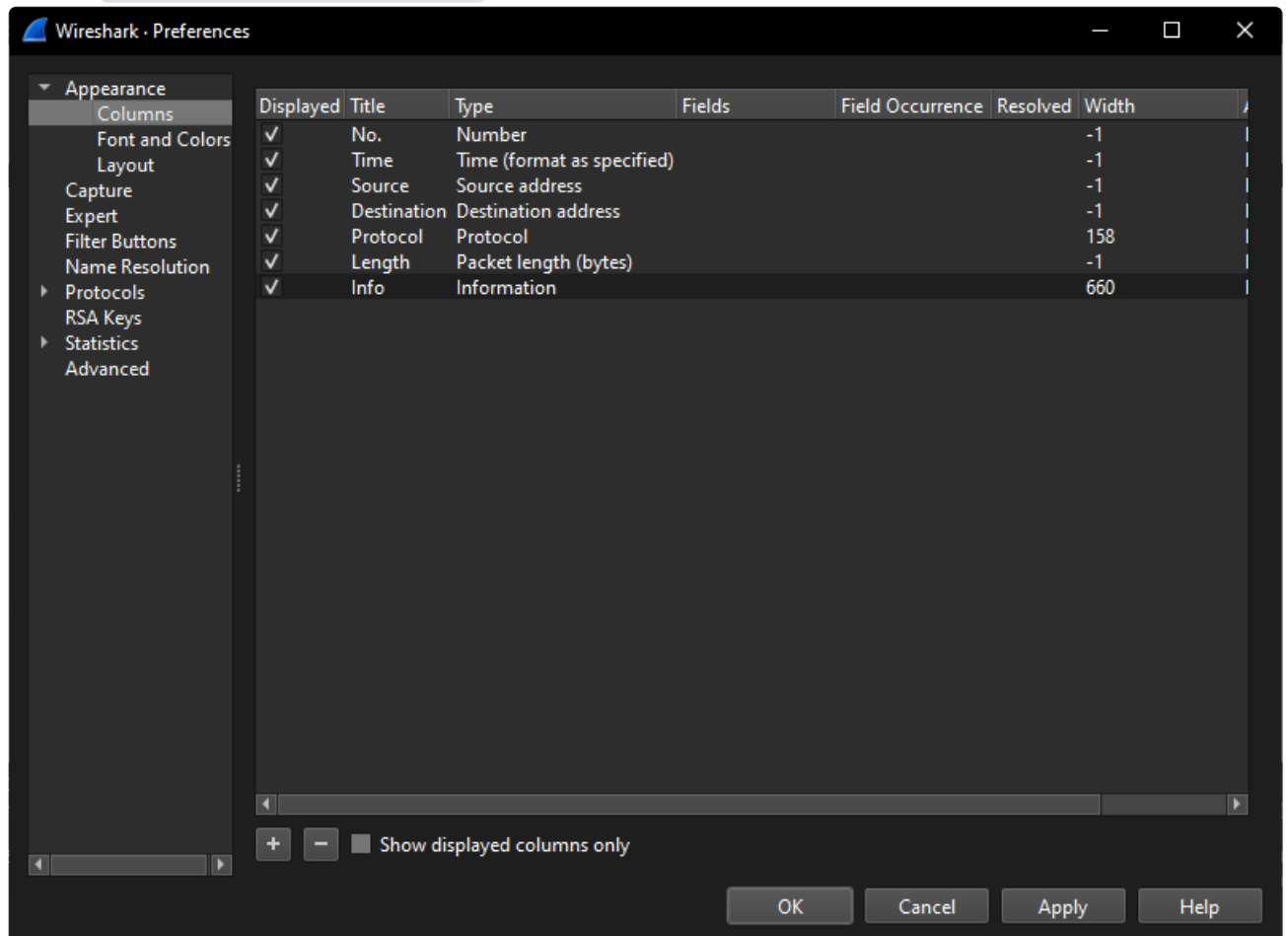


Add a custom column to the wireshark packet list pane for better visibility.

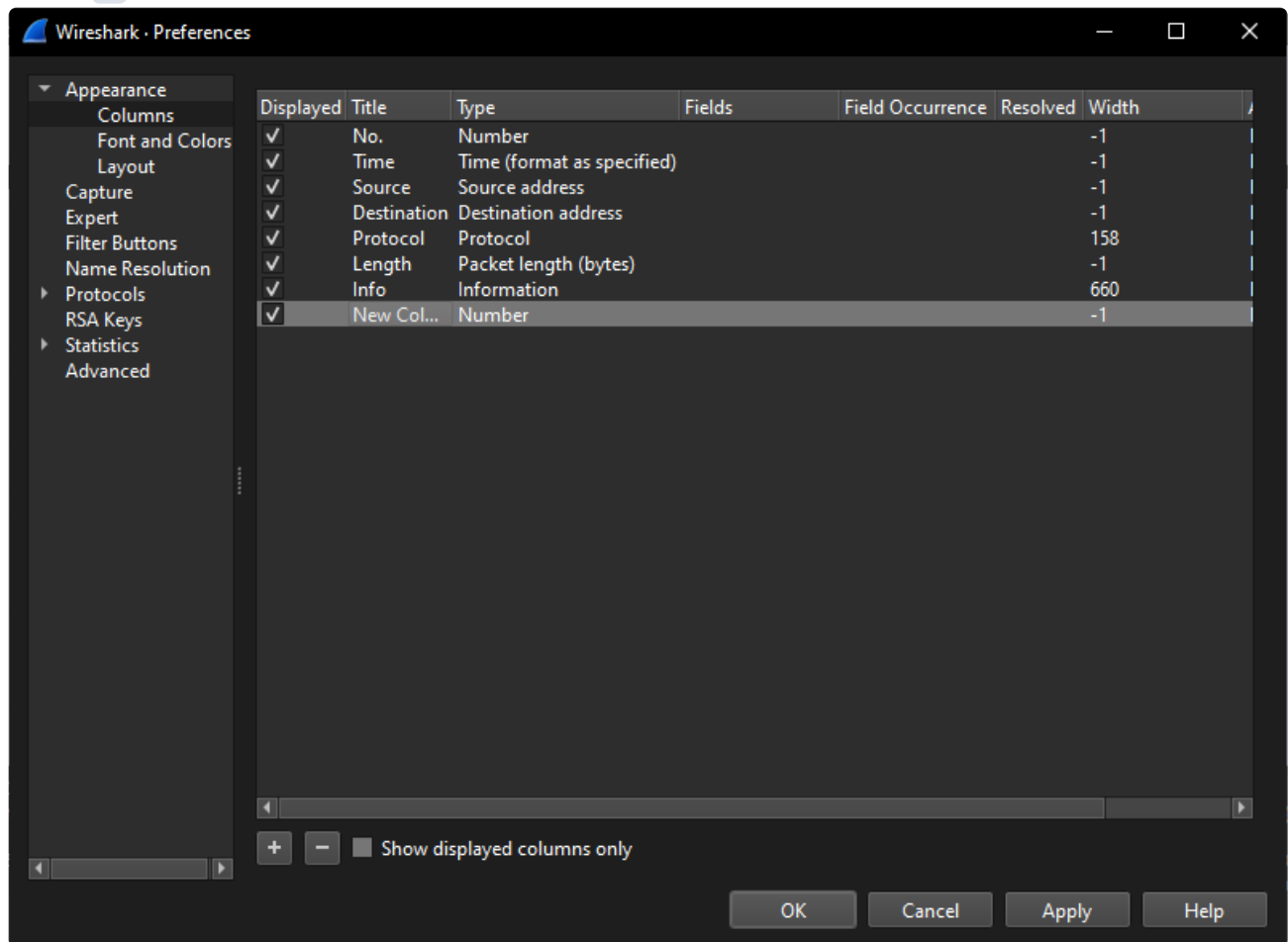
1. Go to **Edit → Preferences**



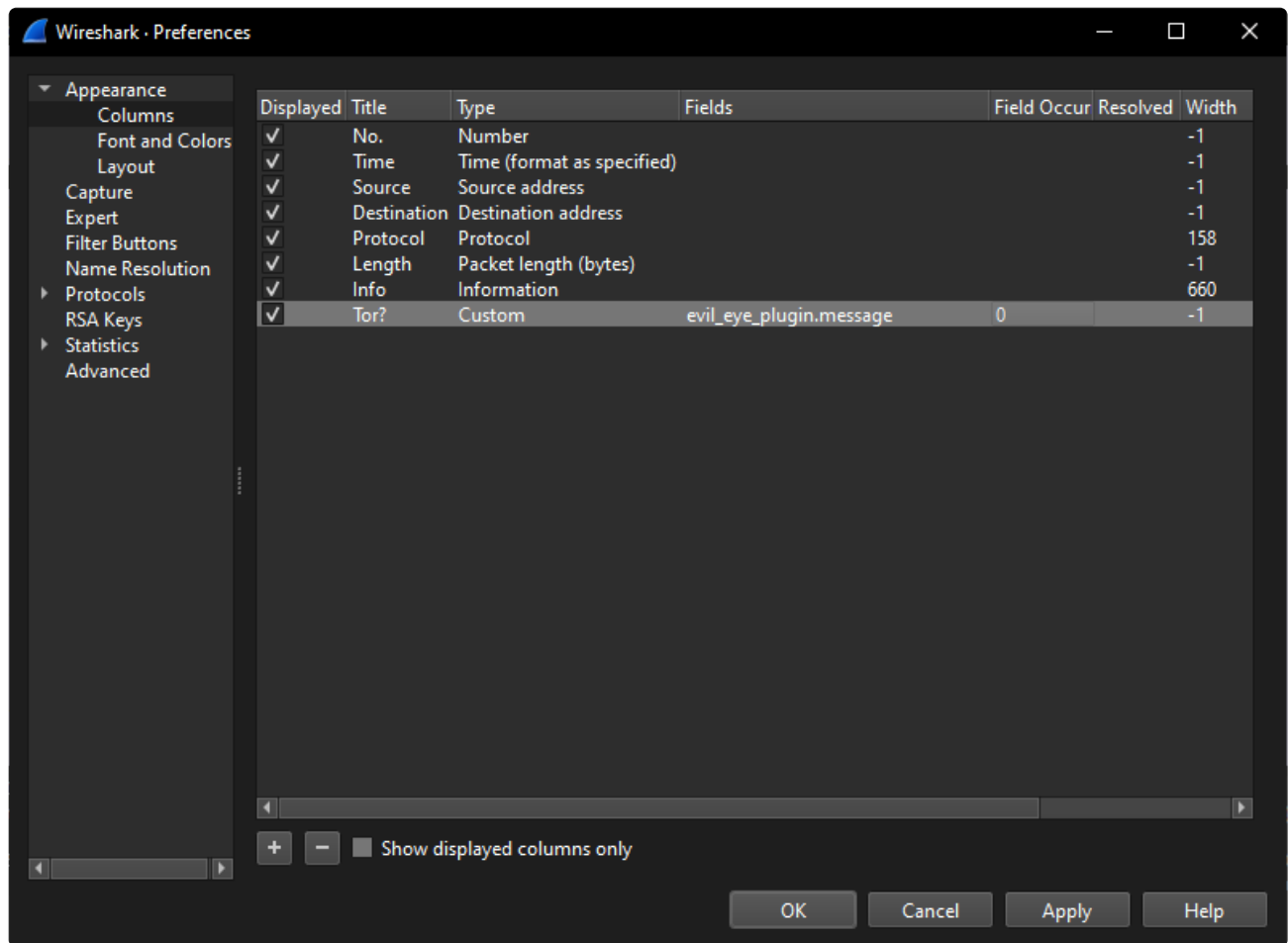
2. Go to **Appearance → Columns**



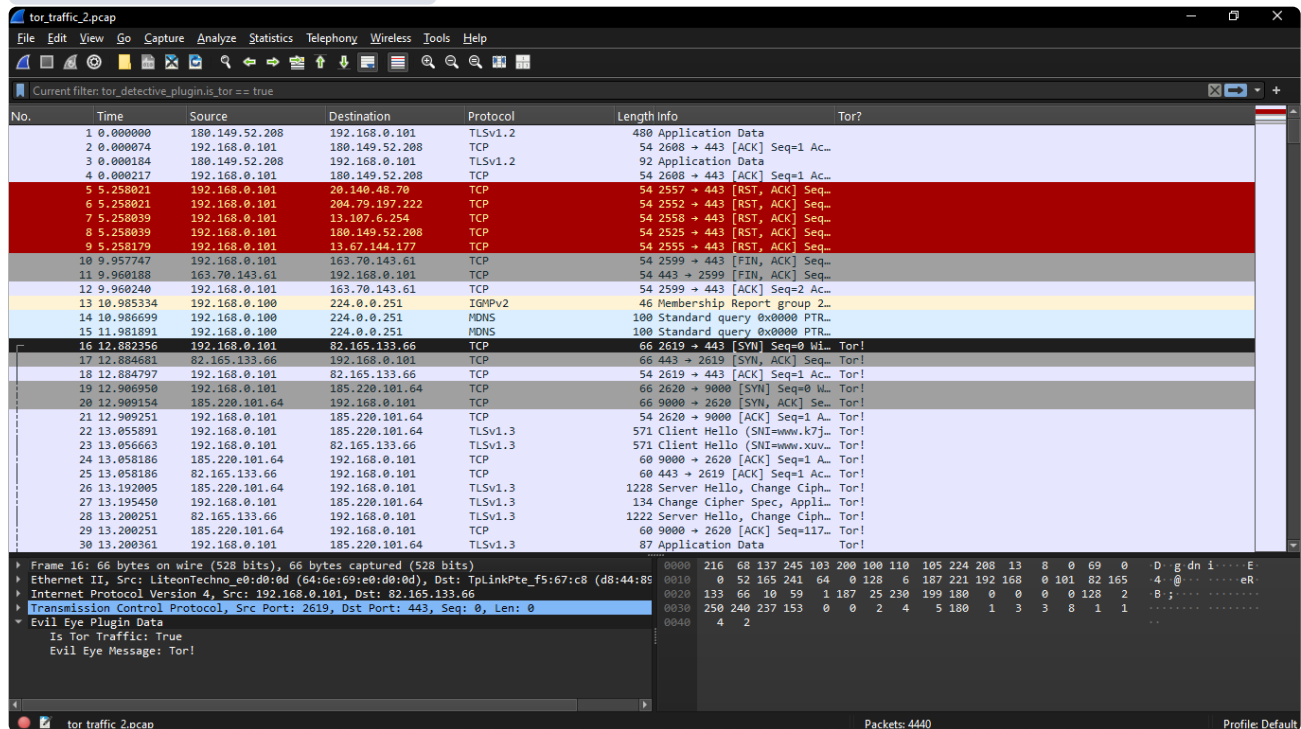
3. Click **+** button. It will add new column.



4. Edit **Title** : "Tor?", **Type** = "Custom" and **Fields** =
"evil_eye_plugin.message"



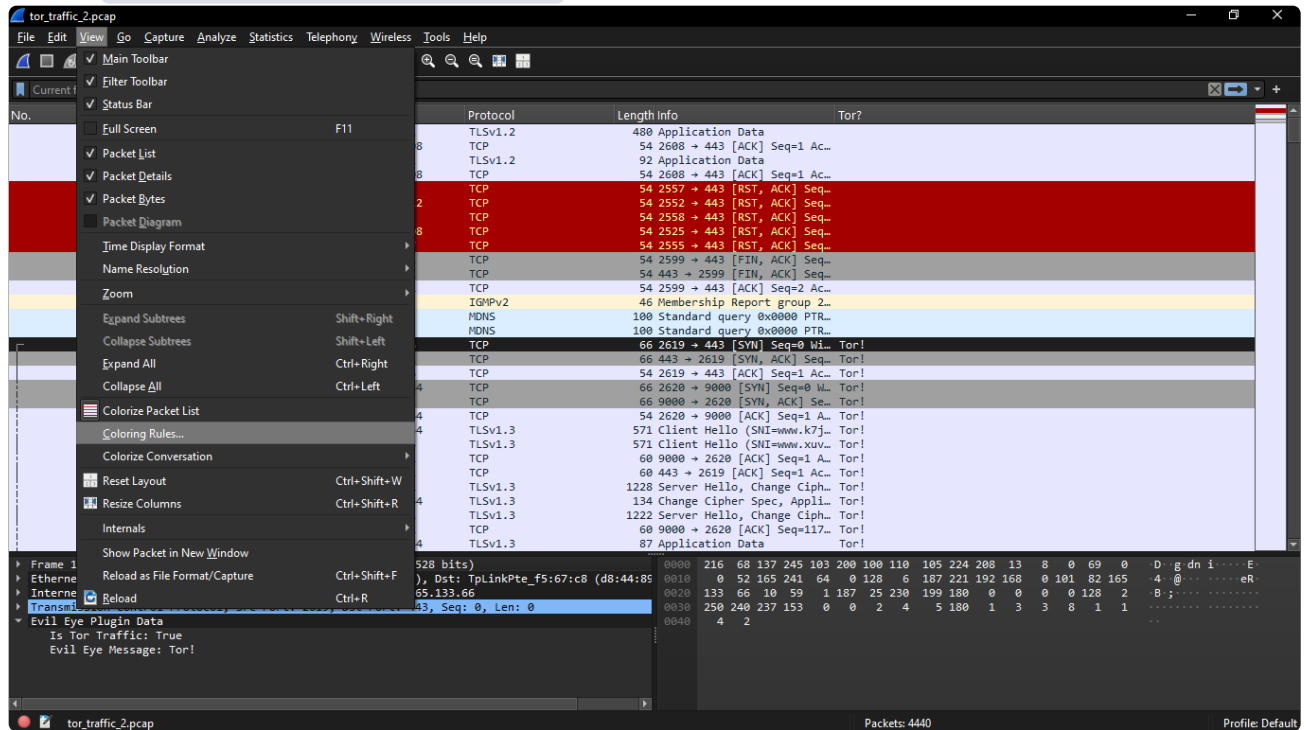
5. **Apply** and **OK**. Plugin will automatically update this column based on **evil_eye_plugin.message** field set by plugin.



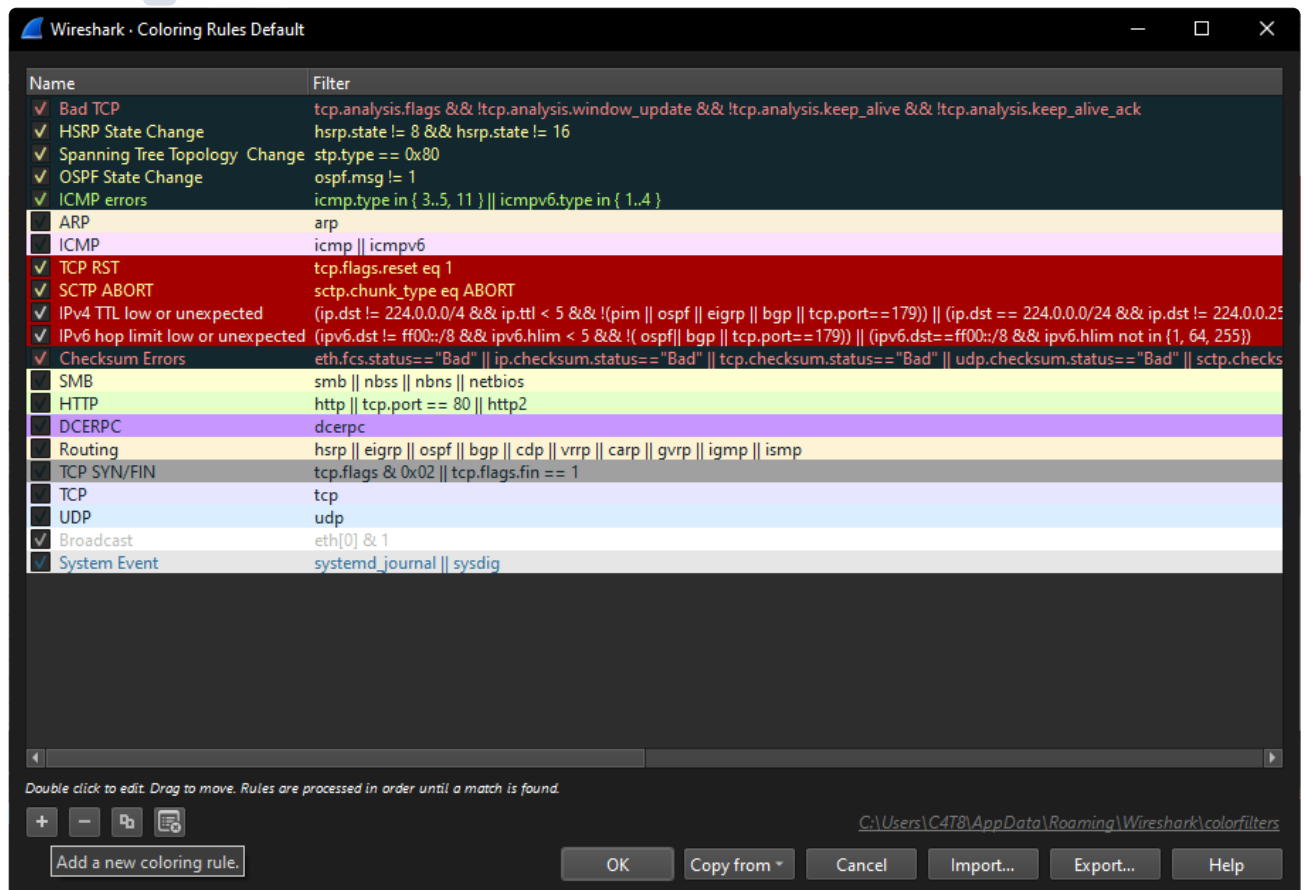
Apply color rules

1. To make this even more stand out, Let's add color rules.

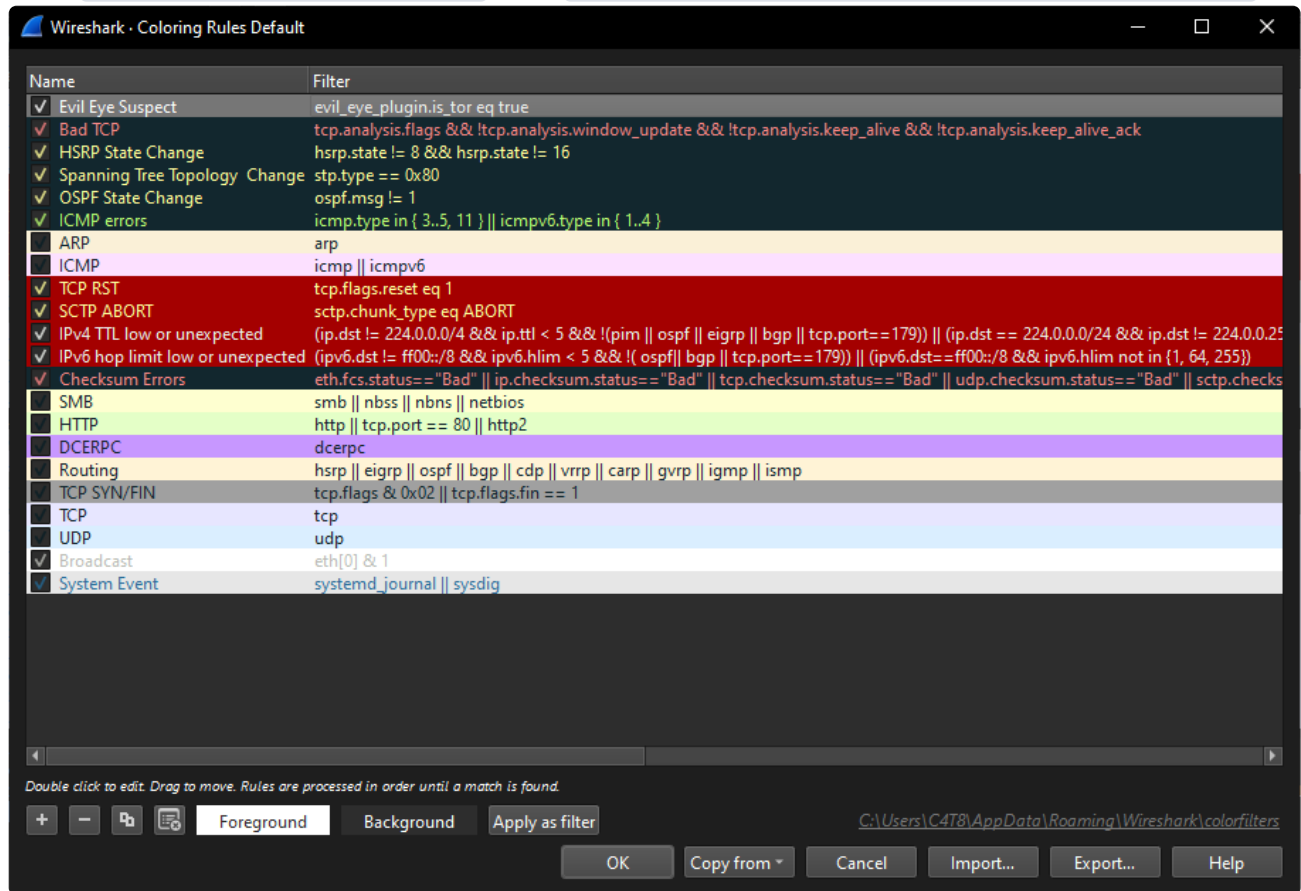
2. Go to **View → Coloring Rules...**



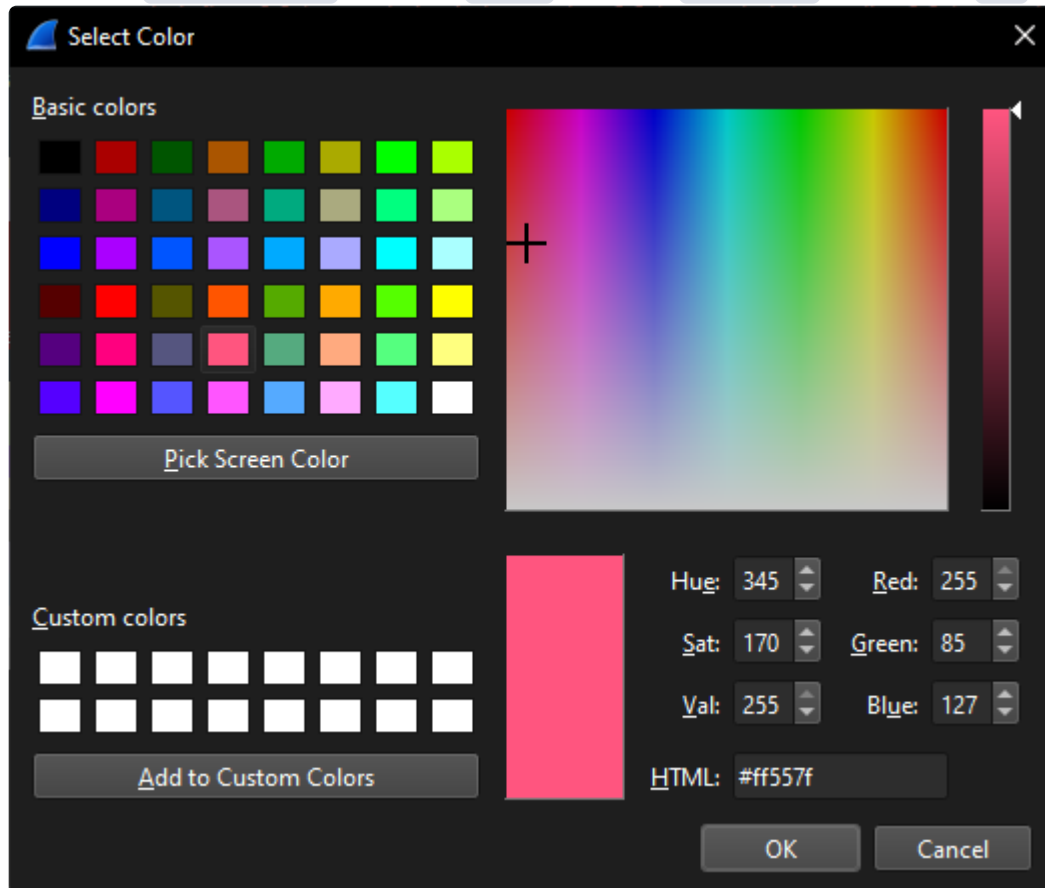
3. Click on **+** to add new rule.



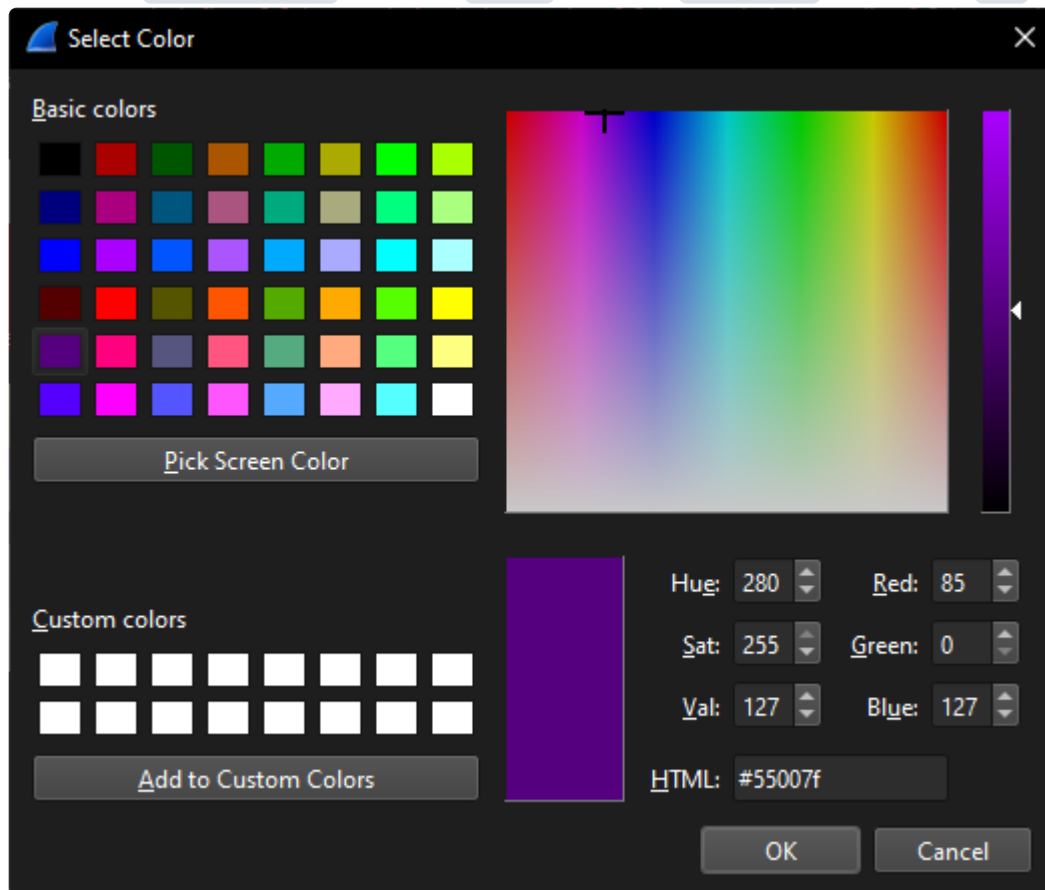
4. Edit Name: Evil Eye Suspect and Filter: evil_eye_plugin.is_tor == true



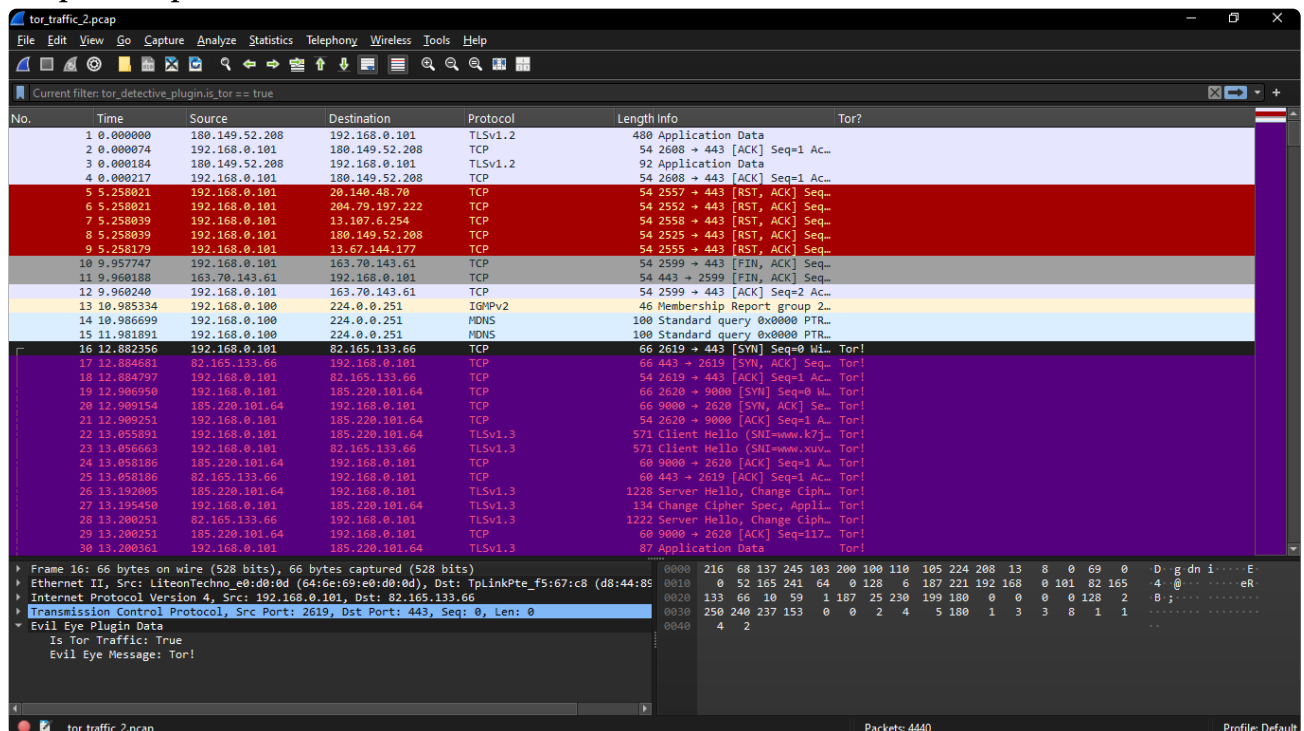
5. Click on Foreground and set HTML field to #ff557f and click OK



6. Click on **Background** and set **HTML** field to **#55007f** and click **OK**

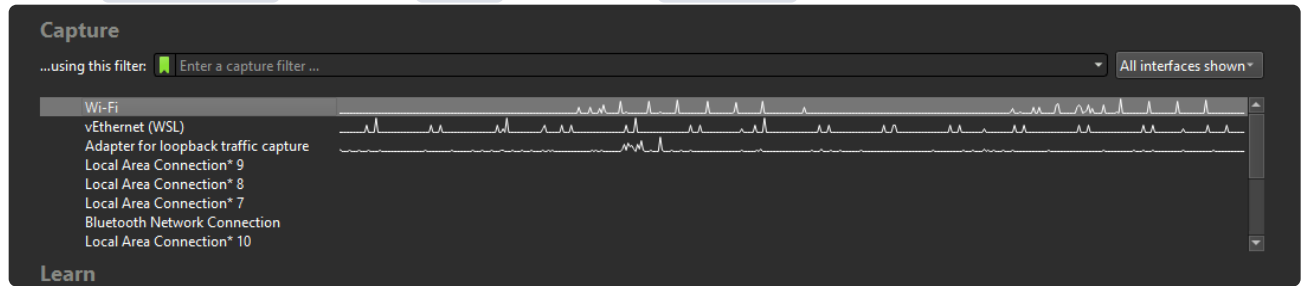


7. Click on **OK**. This will be applied to all the packets for which **evil_eye_plugin.is_tor** field is **true**. This field will be set **true** by plugin for suspected packets.

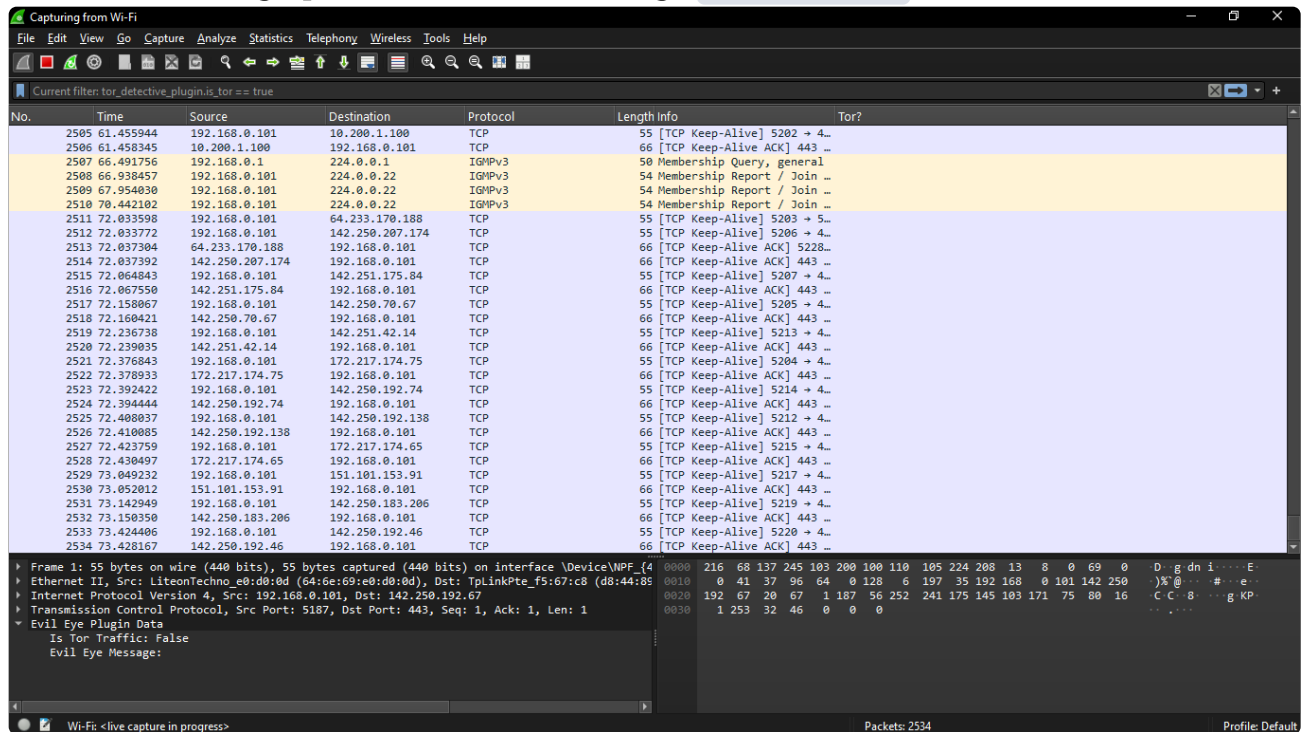


Working on WiFi Traffic

1. Let's check on wifi traffic.
2. Open **Wireshark** click on **WiFi** and then **Capture**.

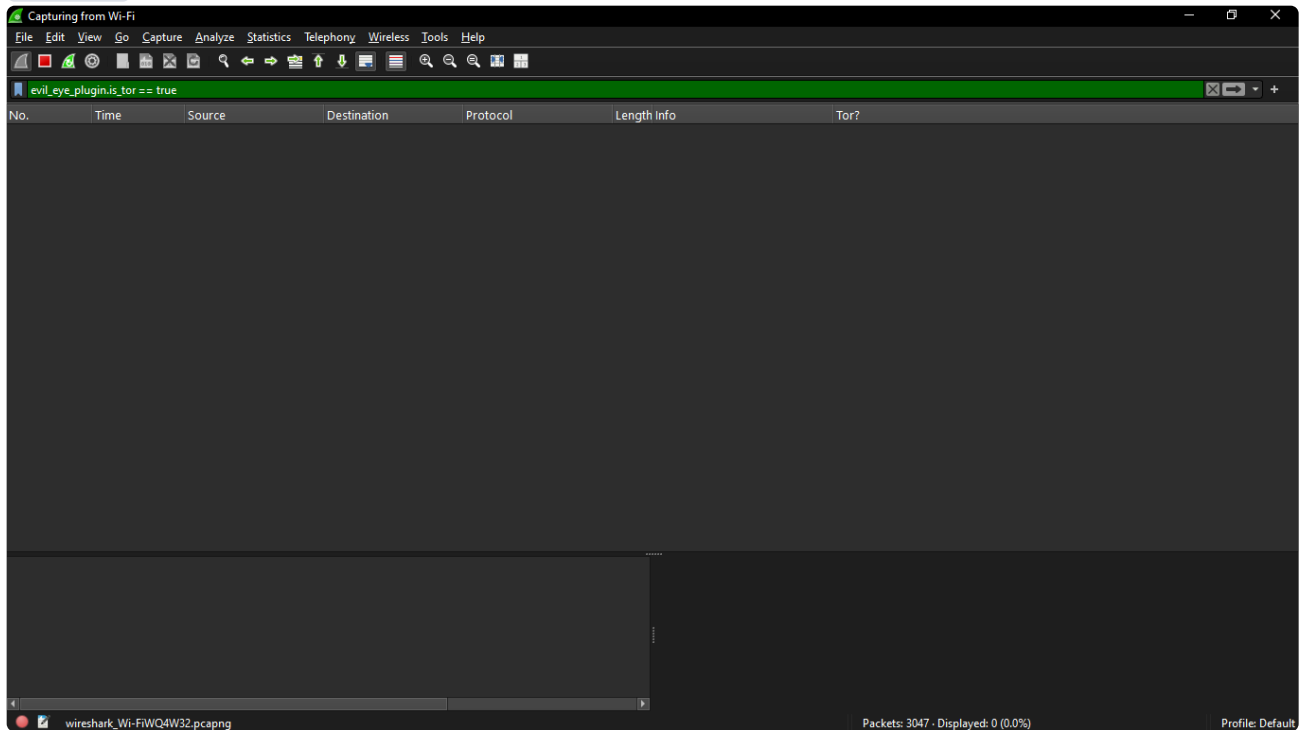


3. Generated enough packets for normal usage **Without Tor**



4. Let's apply filter **evil_eye_plugin.is_tor eq true** and see how many packets we have captured. **This is non tor traffic. So it should not capture any**

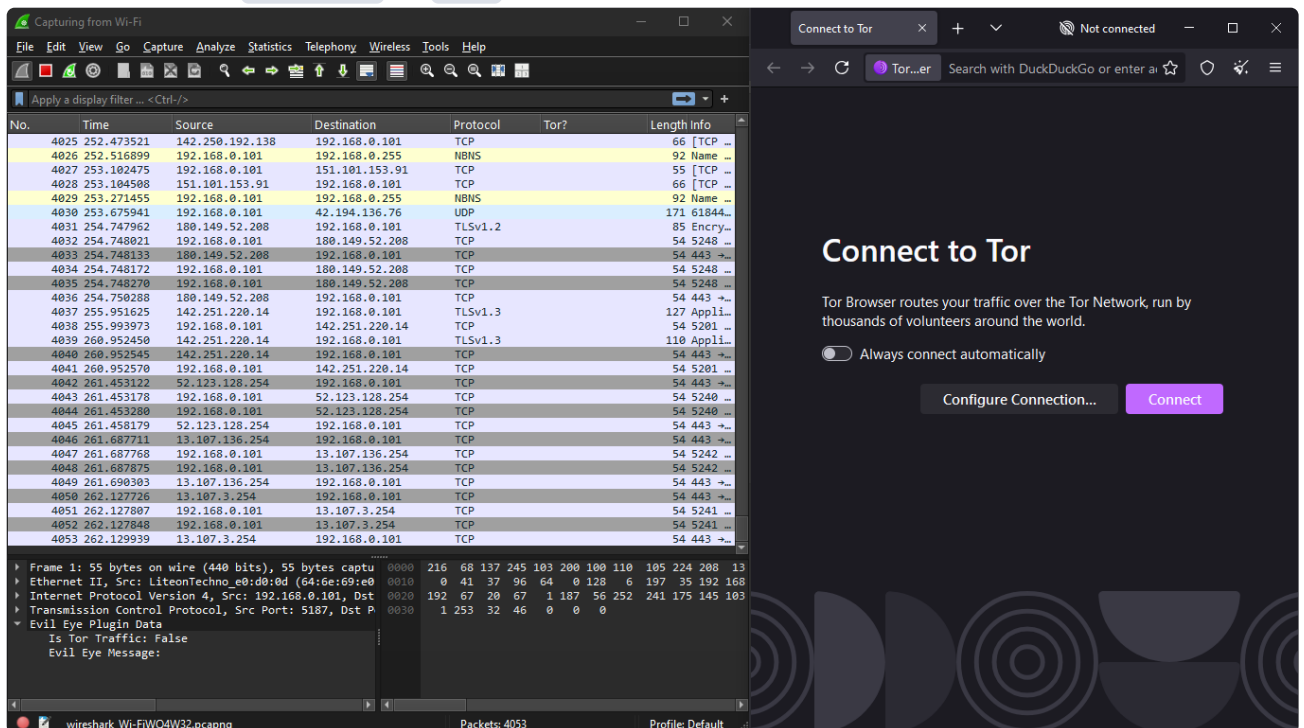
packet.



5. So far, so good.

6. Let's open **Tor** browser and see if it manages to detect accurately tor traffic.

7. Before clicking **Connect** in **Tor**



8. After clicking **Connect** in **Tor**

The screenshot displays two windows side-by-side. The left window is Wireshark, showing a packet capture from a Wi-Fi interface. The right window is a web browser displaying the Tor network connection status.

Wireshark Packet Capture:

No.	Time	Source	Destination	Protocol	Tor?	Length	Info
4278	301.212227	185.220.101.64	192.168.0.101	TCP	Tor!	60	9000 →
4279	301.276492	45.145.164.86	192.168.0.101	TLSv1.3	Tor!	1126	Appli...
4280	301.276899	192.168.0.101	45.145.164.86	TLSv1.3	Tor!	1104	Appli...
4281	301.291049	45.145.164.86	192.168.0.101	TCP	Tor!	60	9000 →
4282	301.291252	185.220.101.64	192.168.0.101	TLSv1.3	Tor!	398	Appli...
4283	301.291874	192.168.0.101	185.220.101.64	TLSv1.3	Tor!	1104	Appli...
4284	301.302676	185.220.101.64	192.168.0.101	TCP	Tor!	60	9000 →
4285	301.423458	185.220.101.64	192.168.0.101	TLSv1.3	Tor!	590	Appli...
4286	301.425626	192.168.0.101	185.220.101.64	TLSv1.3	Tor!	590	Appli...
4287	301.427792	185.220.101.64	192.168.0.101	TCP	Tor!	60	9000 →
4288	301.456187	TpLinkPte_f5:67:c8	LiteonTechno_e0:d0:...	ARP		42	Who h...
4289	301.456211	LiteonTechno_e0:d0:...	TpLinkPte_f5:67:c8	ARP		42	192.1...
4290	301.6412380	185.220.101.64	192.168.0.101	TLSv1.3	Tor!	590	Appli...
4291	301.644647	45.145.164.86	192.168.0.101	TLSv1.3	Tor!	590	Appli...
4292	301.647175	192.168.0.101	45.145.164.86	TLSv1.3	Tor!	590	Appli...
4293	301.649918	45.145.164.86	192.168.0.101	TCP	Tor!	60	9000 →
4294	301.661942	192.168.0.101	185.220.101.64	TCP	Tor!	54	5268 →
4295	301.962119	45.145.164.86	192.168.0.101	TLSv1.3	Tor!	590	Appli...
4296	302.002160	192.168.0.101	45.145.164.86	TLSv1.3	Tor!	590	Appli...
4297	302.005210	45.145.164.86	192.168.0.101	TCP	Tor!	60	9000 →
4298	302.005248	192.168.0.101	45.145.164.86	TLSv1.3	Tor!	590	Appli...
4299	302.008243	45.145.164.86	192.168.0.101	TCP	Tor!	60	9000 →
4300	302.130463	45.145.164.86	192.168.0.101	TLSv1.3	Tor!	590	Appli...
4301	302.131544	192.168.0.101	45.145.164.86	TLSv1.3	Tor!	590	Appli...
4302	302.137739	45.145.164.86	192.168.0.101	TCP	Tor!	60	9000 →
4303	302.293754	45.145.164.86	192.168.0.101	TLSv1.3	Tor!	590	Appli...
4304	302.295426	192.168.0.101	45.145.164.86	TLSv1.3	Tor!	590	Appli...
4305	302.297846	45.145.164.86	192.168.0.101	TCP	Tor!	60	9000 →
4306	302.358256	192.168.0.101	185.125.190.56	NTP		90	NTP V...

Browser Window:

The browser window shows the Tor network connection status. The address bar displays "Search with DuckDuckGo or enter address". The main content area features the Tor logo and the text "Explore. Privately." Below this, there is a search bar with the text "Search with DuckDuckGo" and a toggle switch for "Onionize". The bottom of the page states "You're ready for the world's most private browsing experience."

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Tor?	Length	Info
6072	342.552108	204.79.197.222	192.168.0.101	TLSv1.2		396	New S...
6073	342.552108	204.79.197.222	192.168.0.101	TLSv1.2		123	Appli...
6074	342.552194	192.168.0.101	204.79.197.222	TCP		54	5282 ...
6075	342.552914	192.168.0.101	204.79.197.222	TLSv1.2		92	Appli...
6076	342.554204	204.79.197.222	192.168.0.101	TLSv1.2		285	Appli...
6077	342.554204	204.79.197.222	192.168.0.101	TLSv1.2		92	Appli...
6078	342.554265	192.168.0.101	204.79.197.222	TCP		54	5282 ...
6079	342.554356	204.79.197.222	192.168.0.101	TCP		60	443 ...
6080	342.559724	192.168.0.101	157.140.154.86	TCP	Tor!	54	5282 ...
6081	342.624549	TpLinkPte fs:67:c8	LiteonTechno e0:d0:0d	ARP		42	Who h...
6082	342.624570	TpLinkPte fs:67:c8	TpLinkPte fs:67:c8	ARP		42	192.1...
6083	342.828484	185.220.101.64	192.168.0.101	TCP	Tor!	1507	9000 ...
6084	342.828484	185.220.101.64	192.168.0.101	TCP	Tor!	1514	9000 ...
6085	342.828480	192.168.0.101	185.220.101.64	TCP	Tor!	54	5268 ...
6086	342.828587	185.220.101.64	192.168.0.101	TLSv1.3	Tor!	761	Appli...
6087	342.828553	192.168.0.101	185.220.101.64	TLSv1.3	Tor!	590	Appli...
6088	342.830789	185.220.101.64	192.168.0.101	TCP	Tor!	60	9000 ...
6089	342.855977	185.220.101.64	192.168.0.101	TCP	Tor!	1507	9000 ...
6090	342.904900	192.168.0.101	185.220.101.64	TCP	Tor!	54	5268 ...
6091	342.972144	185.220.101.64	192.168.0.101	TLSv1.3	Tor!	979	Appli...
6092	343.022683	192.168.0.101	185.220.101.64	TCP	Tor!	54	5268 ...
6093	343.096567	185.220.101.64	192.168.0.101	TCP	Tor!	1507	9000 ...
6094	343.096764	185.220.101.64	192.168.0.101	TCP	Tor!	1507	9000 ...
6095	343.096803	192.168.0.101	185.220.101.64	TCP	Tor!	54	5268 ...
6096	343.109500	185.220.101.64	192.168.0.101	TLSv1.3	Tor!	768	Appli...
6097	343.116768	185.220.101.64	192.168.0.101	TCP	Tor!	1507	9000 ...
6098	343.116837	192.168.0.101	185.220.101.64	TCP	Tor!	54	5268 ...
6099	343.249153	185.220.101.64	192.168.0.101	TLSv1.3	Tor!	165	Appli...
6100	343.306729	192.168.0.101	185.220.101.64	TCP	Tor!	54	5268 ...

Frame 1: 55 bytes on wire (440 bits), 55 bytes captured on interface \Device\NPF...
Ethernet II, Src: LiteonTechno_e0:d0:0d (64:6e:69:e0), Dst: 192.168.0.101, Protocol: Internet Protocol Version 4, Src: 192.168.0.101, Dst: 192.168.0.101, Length: 103
Transmission Control Protocol, Src Port: 5187, Dst Port: 5268, Seq: 1, Ack: 518, Len: 0
Evil Eye Plugin Data
Is Tor Traffic: False
Evil Eye Message:

wireshark_Wi-FWQ4W32.pcapng Packets: 6100 Profile: Default

onion browser at DuckDuckGo

https://duckduckgo.com/?q=onion+bro

DuckDuckGo

onion browser

All Images Videos News Maps Shopping

Always private Region Safe search Date

torproject.org
https://www.torproject.org/download

Download - Tor Project

We encourage you to try Onion Browser. Our mission: To create and deploy free and open source anonymity at unrestricted availability and use, and furthering their scientific research.

Download Tor

Available in 36 Different Languages

AR

Support

Success

DOWNLOADING

*Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

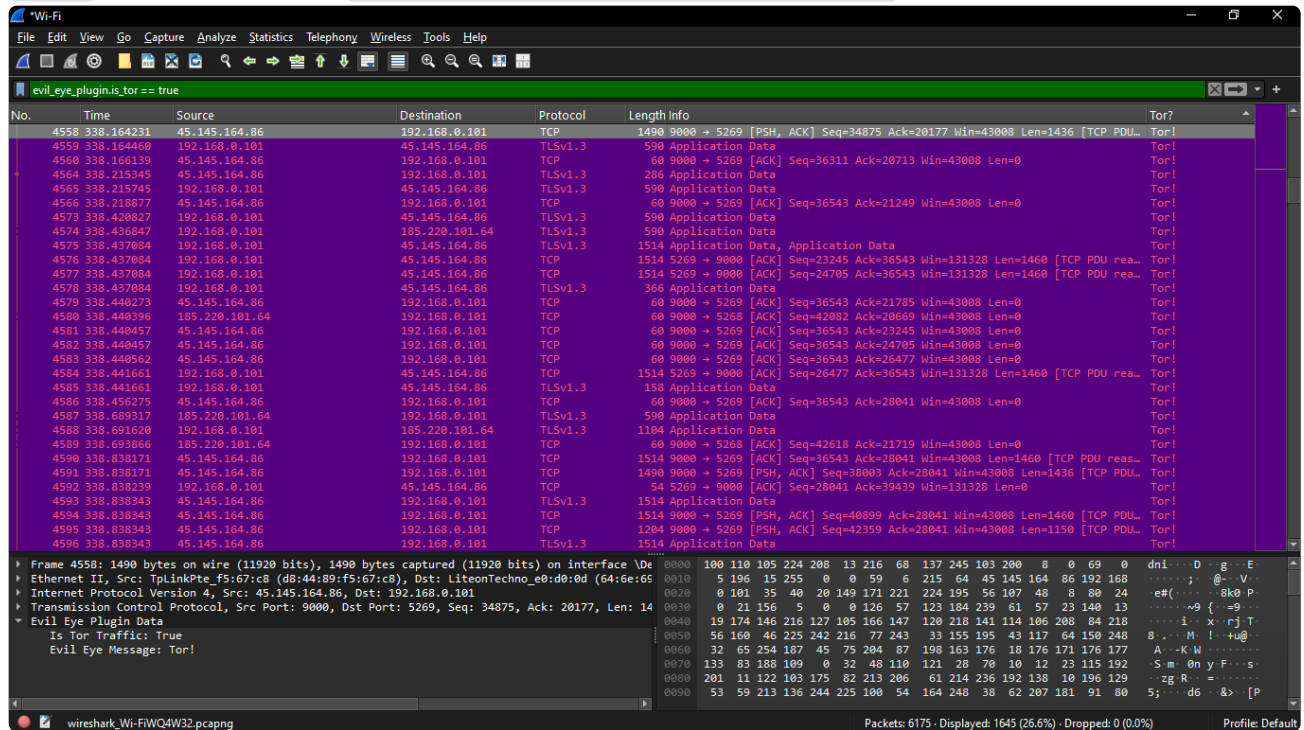
Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info	Tor?
6161	347.670139	23.55.108.18	192.168.0.101	TCP	60	443 → 5284 [ACK] Seq=4069 Ack=541 Win=43008 Len=0	
6162	347.702494	23.55.108.18	192.168.0.101	TLSv1.2	122	Application Data	
6163	347.702583	23.55.108.18	192.168.0.101	TLSv1.2	92	Application Data	
6164	347.702609	192.168.0.101	23.55.108.18	TCP	54	5284 → 443 [ACK] Seq=541 Ack=4175 Win=130816 Len=0	
6165	347.702718	23.55.108.18	192.168.0.101	TLSv1.2	207	Application Data	
6166	347.702718	23.55.108.18	192.168.0.101	TLSv1.2	92	Application Data	
6167	347.702733	192.168.0.101	23.55.108.18	TLSv1.2	92	Application Data	
6168	347.702700	192.168.0.101	23.55.108.18	TCP	54	5284 → 443 [ACK] Seq=579 Ack=4366 Win=130816 Len=0	
6169	347.702802	192.168.0.101	23.55.108.18	TCP	54	5284 → 443 [FIN, ACK] Seq=579 Ack=4366 Win=130816 Len=0	
6170	347.704659	23.55.108.18	192.168.0.101	TCP	54	443 → 5284 [ACK] Seq=4366 Ack=579 Win=43008 Len=0	
6171	347.705065	23.55.108.18	192.168.0.101	TCP	54	443 → 5284 [FIN, ACK] Seq=4366 Ack=580 Win=43008 Len=0	
6172	347.705095	192.168.0.101	23.55.108.18	TCP	54	5284 → 443 [ACK] Seq=580 Ack=4367 Win=130816 Len=0	
6173	348.830355	192.168.0.101	13.107.246.68	TCP	54	[TCP Retransmission] 5255 → 443 [FIN, ACK] Seq=887 Ack=8440 Win=262144...	
6175	350.919056	192.168.0.101	45.145.164.86	TCP	54	5269 → 9000 [ACK] Seq=77875 Ack=356409 Win=130816 Len=0	Tor!
6180	296.620490	192.168.0.101	185.220.101.64	TLSv1.3	571	Client Hello (SNI=www.nixwe3qmw5r4s6thotan.com)	Tor!
6181	296.621302	192.168.0.101	45.145.164.86	TLSv1.3	571	Client Hello (SNI=www.57zwha5ea6brbfljyjqx.com)	Tor!
6182	296.622712	185.220.101.64	192.168.0.101	TCP	60	9000 → 5268 [ACK] Seq=1 Ack=518 Win=43008 Len=0	Tor!
6183	296.623183	45.145.164.86	192.168.0.101	TCP	60	9000 → 5269 [ACK] Seq=1 Ack=518 Win=43008 Len=0	Tor!
6184	296.749281	45.145.164.86	192.168.0.101	TLSv1.3	1225	Server Hello, Change Cipher Spec, Application Data, Application Data, ...	Tor!
6185	296.751850	192.168.0.101	45.145.164.86	TLSv1.3	134	Change Cipher Spec, Application Data	Tor!
6186	296.753920	45.145.164.86	192.168.0.101	TCP	60	9000 → 5269 [ACK] Seq=1172 Ack=598 Win=43008 Len=0	Tor!
6187	296.753965	192.168.0.101	45.145.164.86	TLSv1.3	87	Application Data	Tor!
6188	296.757749	45.145.164.86	192.168.0.101	TCP	60	9000 → 5269 [ACK] Seq=1172 Ack=631 Win=43008 Len=0	Tor!
6189	296.767624	185.220.101.64	192.168.0.101	TLSv1.3	1217	Server Hello, Change Cipher Spec, Application Data, Application Data, ...	Tor!
6190	296.769774	192.168.0.101	185.220.101.64	TLSv1.3	134	Change Cipher Spec, Application Data	Tor!
6191	296.772353	185.220.101.64	192.168.0.101	TCP	60	9000 → 5268 [ACK] Seq=1164 Ack=598 Win=43008 Len=0	Tor!
6192	296.772406	192.168.0.101	185.220.101.64	TLSv1.3	87	Application Data	Tor!
6193	296.774256	185.220.101.64	192.168.0.101	TCP	60	9000 → 5268 [ACK] Seq=1164 Ack=631 Win=43008 Len=0	Tor!
6194	296.870990	45.145.164.86	192.168.0.101	TLSv1.3	133	Application Data	Tor!
6195	296.875602	45.145.164.86	192.168.0.101	TLSv1.3	1502	Application Data	Tor!

Frame 4162: 60 bytes on wire (480 bits), 60 bytes captured on interface \Device\NPF...
Ethernet II, Src: TpLinkPte fs:67:c8 (d8:44:89:f5:67:c8), Dst: LiteonTechno_e0:d0:0d (64:6e:69:e0), Protocol: Internet Protocol Version 4, Src: tor-exit-64.cccs.de (185.220.101.64), Dst: 192.168.0.101 (192.168.0.101), Length: 115
Transmission Control Protocol, Src Port: 9000, Dst Port: 5268, Seq: 1, Ack: 518, Len: 0
Evil Eye Plugin Data
Is Tor Traffic: True
Evil Eye Message: Tor!

wireshark_Wi-FWQ4W32.pcapng Packets: 6175 - Dropped: 0 (0.0%) Profile: Default

9. Stop and apply filter `evil_eye_plugin.is_tor == true`

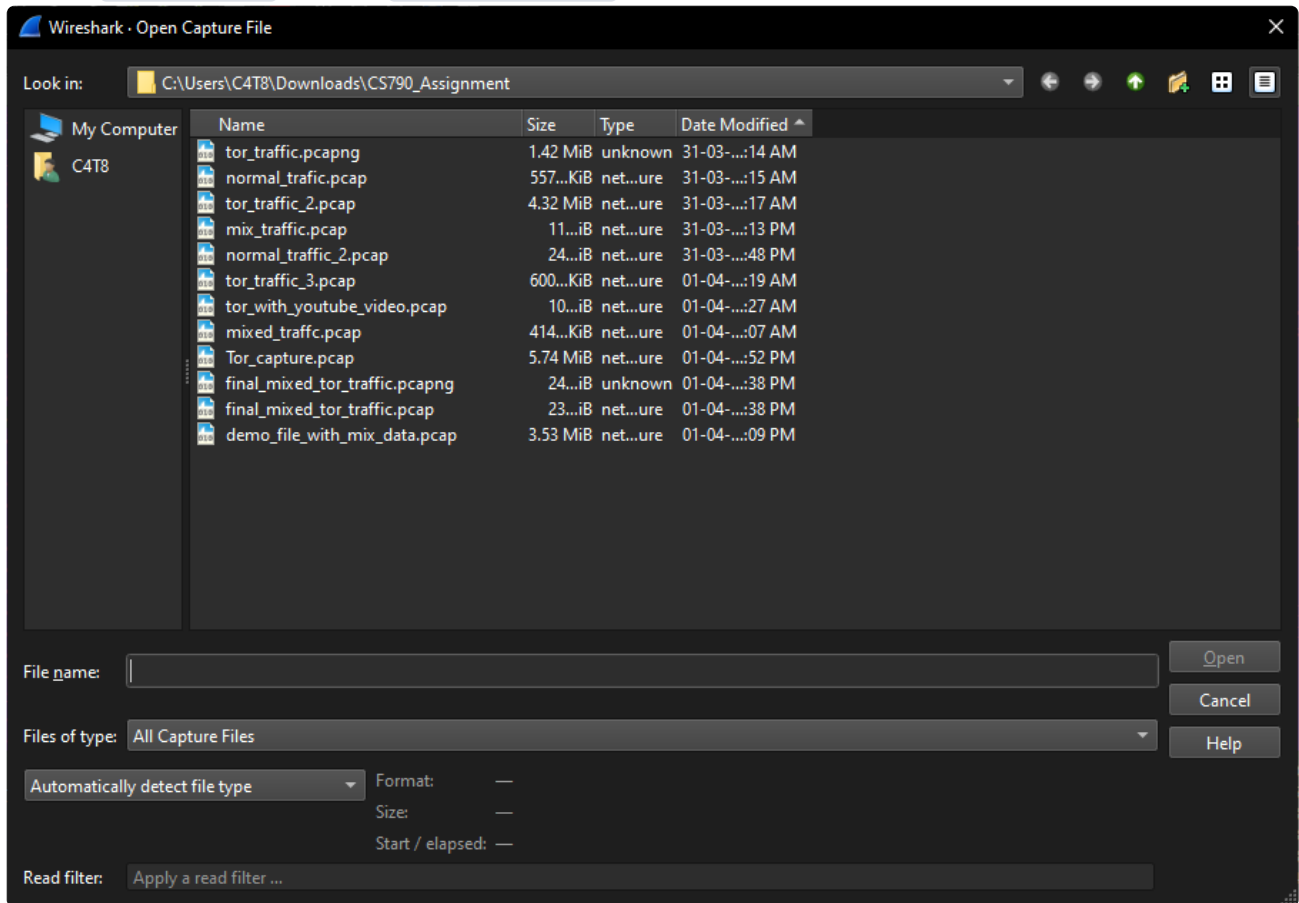


10. As we can see.. it successfully suspects all correct **TOR** traffic while minimizing false positive to **0**.

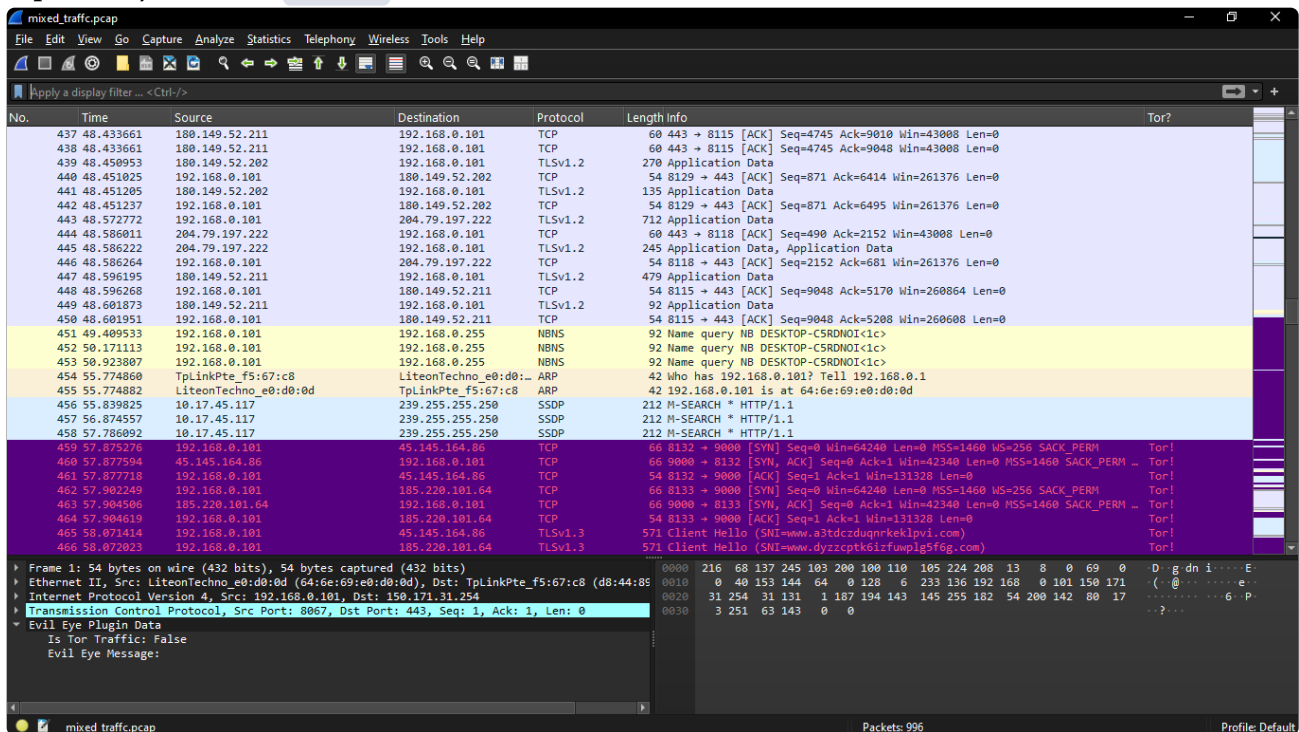
Working on PCAP file traffic

1. This plugin will work on **PCAP** files similarly.

2. Open Wireshark . Go to File → Open



3. Open any desired .pcap file.



4. As we can see plugin is working as expected of .pcap files.